

Question 1.

Marks

(a) Find $\int_0^{\frac{\pi}{2}} (\cos^3 x) \cdot \sqrt{\sin x} \, dx$. 3

(b) Find $\int \frac{dx}{1-e^x}$. 2

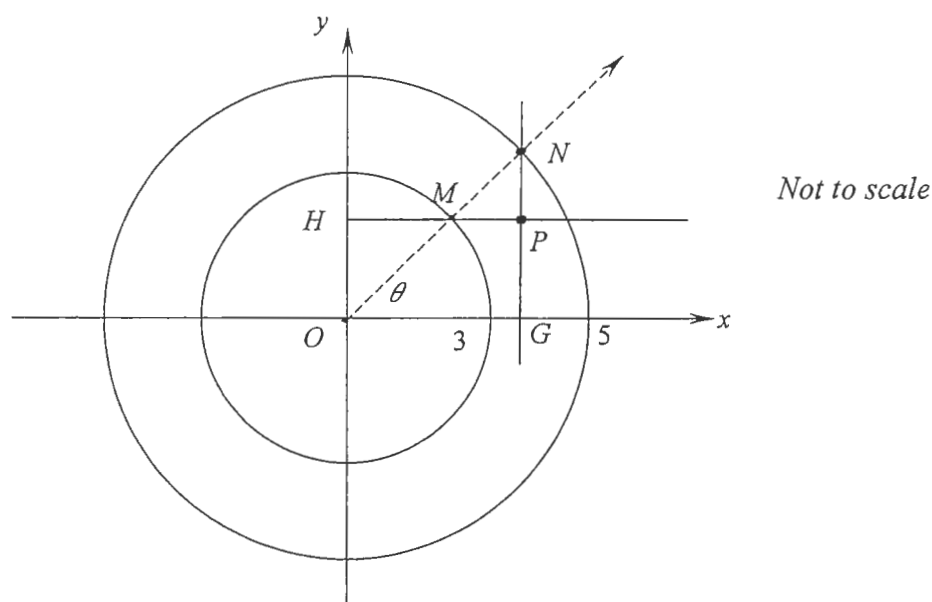
(c) (i) Find the real numbers a and b such that: 2

$$\frac{x-1}{(x-2)(x^2-4x+5)} = \frac{1}{x-2} + \frac{ax+b}{x^2-4x+5}.$$

(ii) Hence, find $\int \frac{x-1}{(x-2)(x^2-4x+5)} \, dx$. 3

(d) Using the substitution $t = \tan \frac{x}{2}$, find $\int \frac{dx}{3+2\cos x}$. 3

(e) 2



The circles above have centres at O and with radii 3 units and 5 units respectively. A ray from O makes an angle of θ with the positive x -axis and cuts the circles at points M and N as shown.

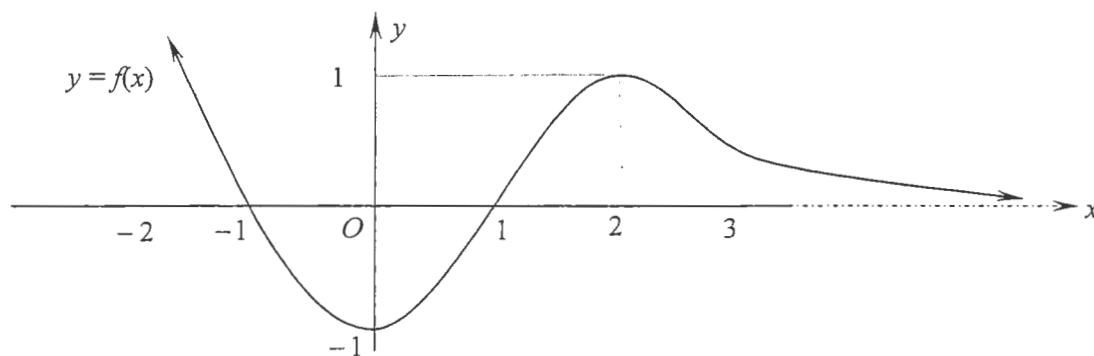
NG is drawn parallel to the y -axis. NG and MH intersect at point P .

Find the Cartesian equation for the locus of P as θ varies.

Question 2. [Start a New Page]

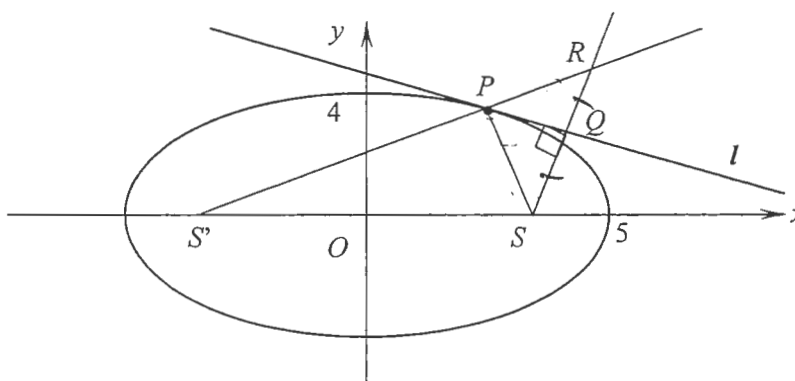
Marks

- (a) The sketch of the graph of the function $y = f(x)$ is shown below.



On separate number planes, sketch the graphs of the following, showing all essential features.

- (i) $y = f(x - 1)$. 2
 - (ii) $y = \frac{1}{f(x)}$. 2
 - (iii) $y = \ln[f(x)]$. 2
 - (iv) $y^2 = f(x)$. 2
 - (v) $y = xf(x)$. 2
- (b) The diagram below shows the ellipse $E: \frac{x^2}{25} + \frac{y^2}{16} = 1$.
 The line l is tangent to the ellipse E at the point P .
 The foci of the ellipse are S and S' . The perpendicular to l through S meets l at the point Q . The lines SQ and $S'P$ meet at the point R .



- (i) Copying the diagram onto your paper and by using the reflection property of the ellipse at P , prove that $SQ = RQ$. 2
- (ii) Explain why $S'R = 10$. 1
- (iii) Hence, or otherwise, prove that Q lies on the circle $x^2 + y^2 = 25$. 2

Question 3. [Start a New Page]

Marks

(a) Let $I = \int_1^3 \frac{\sin^2\left(\frac{\pi x}{8}\right)}{x(4-x)} dx,$

(i) By using the substitution $t = 4 - x$, show that: $I = \int_1^3 \frac{\cos^2\left(\frac{\pi t}{8}\right)}{t(4-t)} dt.$ 2

(ii) Hence, find the value of I . 3

(b) The hyperbola H has the Cartesian equation $\frac{x^2}{4} - \frac{y^2}{5} = 1$,
where $P(2\sec\theta, \sqrt{5}\tan\theta)$ is a point on H .

(i) Find the eccentricity e . 1

(ii) Find the coordinates of the foci. 1

(iii) State the equation of the asymptotes. 1

(iv) Sketch the hyperbola H . 2

(v) Show that the tangent to H at P has the equation: 2

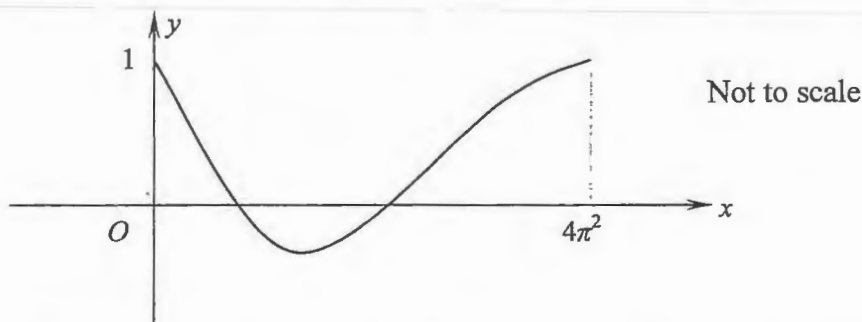
$$\frac{x \sec \theta}{2} - \frac{y \tan \theta}{\sqrt{5}} = 1.$$

(vi) If this tangent cuts the asymptotes at L and M , prove that $LP = PM$. 2

(c) For the general ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, for $a > b$, describe the effect on the ellipse as the eccentricity $e \rightarrow 0^+$. 1

- (a) Show that the equation of the normal to the ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where the eccentricity $e = \frac{1}{\sqrt{3}}$ at point $P(a \cos \theta, b \sin \theta)$ is $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = \frac{a^2}{3}$. 3

- (b) (i) Given the sketch of the curve $y = \cos \sqrt{x}$, for $0 \leq x \leq 4\pi^2$, 1



Show that the area, A square units, bounded by the curve $y = \cos(\sqrt{|x|})$ and the x -axis for $-\frac{\pi^2}{4} \leq x \leq \frac{\pi^2}{4}$, can be expressed as

$$A = 2 \int_0^{\frac{\pi^2}{4}} \cos \sqrt{x} \, dx.$$

- (ii) Hence, find the area A . 4

- (c) Given $I_n = \int_1^e (\ln x)^n \, dx$, for $n = 0, 1, 2, \dots$

- (i) Show that $I_n = e - nI_{n-1}$, for $n = 1, 2, 3, \dots$ 2

- (ii) Let $J_n = \frac{I_n}{n!}$, show that: $\frac{1}{e}(1 + J_{10}) = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{1}{10!}$. 3

- (iii) Given that: $\sum_{r=0}^n \frac{(-1)^r}{r!} = \frac{1}{e} [1 + (-1)^n J_n]$ for $n = 0, 1, 2, \dots$ 2

Deduce the sum to infinity of the series:

$$\sum_{r=0}^{\infty} \frac{(-1)^r}{r!} = 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots, \text{ justify your answer.}$$

THE END



MATHEMATICS Extension 1: Question.....

Suggested Solutions	Marks	Marker's Comments
(c) $\int_0^{\pi/2} \cos^3 x \sqrt{\sin x} dx$ $= \int_0^{\pi/2} (1 - \sin^2 x) \cos x (\sin x)^{1/2} dx$ $= \int_0^{\pi/2} \left[(\sin x)^{1/2} - (\sin x)^{5/2} \right] \cos x dx$ $= \left[\frac{2}{3} (\sin x)^{3/2} - \frac{2}{7} (\sin x)^{7/2} \right]_0^{\pi/2}$ $= \frac{2}{3} - \frac{2}{7} - 0 - 0$ $= \frac{8}{21}$	1 1 1	$\sin^{3/2} x \sin^{7/2} x$ $\sin^{5/2} x$ no marks deducted but not acceptable notation -1 each mistake. some people could <u>Not</u> do $\int x^{5/2} dx$ correctly.
d) $\int \frac{dx}{1-e^x} = \int \frac{-e^x}{1-e^x} dx$ $= -\ln 1-e^x + C$ OR $x - \ln 1-e^x + C$	1 1 1	$\frac{1}{2}$ not 1/1 $\frac{1}{2}$ not simplifying $\ln(e^x) = x$
e) $\frac{x-1}{(x-2)(x^2-4x+5)} = \frac{1}{x-2} + \frac{ax+b}{x^2-4x+5}$ $x-1 = x^2-4x+5 + (ax+b)(x-2)$ $x=0$ $-1 = 5 - 2b$ $b=3$ $x=1$ $0 = 2 - (a+3)$ $a=-1$ $\int \frac{(x-1)}{(x-2)(x^2-4x+5)} dx = \int \frac{1}{x-2} + \frac{3-x}{x^2-4x+5} dx$ $= \int \left(\frac{1}{x-2} - \frac{1}{2} \left(\frac{2x-4}{x^2-4x+5} \right) + \frac{1}{x^2-4x+5} \right) dx$ $= \ln x-2 - \frac{1}{2} \ln(x^2-4x+5) + \tan^{-1}(x-2) + C$	1 1 3	Many found a, b by equating coefficients OK. 1 each expression - difficulty found in balancing coefficients -1/2 for sign error

MATHEMATICS Extension 1 : Question.....

Suggested Solutions	Marks	Marker's Comments
(d) $\int \frac{dx}{3+2\ln x}$ $x = \tan \frac{t}{2}$ $= \int \frac{1}{3 + 2 \ln \left(\frac{1-t^2}{1+t^2} \right)} \cdot \frac{2 dt}{1+t^2}$ $= \int \frac{2 dt}{5+t^2}$ $= \frac{2}{\sqrt{5}} \tan^{-1} \frac{t}{\sqrt{5}} + C$ $= \frac{2}{\sqrt{5}} \tan^{-1} \left(\frac{\tan \frac{t}{2}}{\sqrt{5}} \right) + C$	/ / / /	✓ 1 each mistake. ✓ 1 Not substituting $t = \tan \frac{t}{2}$ some people did $\int \frac{2 dt}{5-t^2}$ ✓ 2 for $\frac{2 dt}{1+t^2}$
e) $P(5 \cos \theta, 3 \sin \theta)$ $x = 5 \cos \theta$ $y = 3 \sin \theta$ $\cos^2 \theta + \sin^2 \theta = 1$ $\frac{x^2}{25} + \frac{y^2}{9} = 1$	/ /	

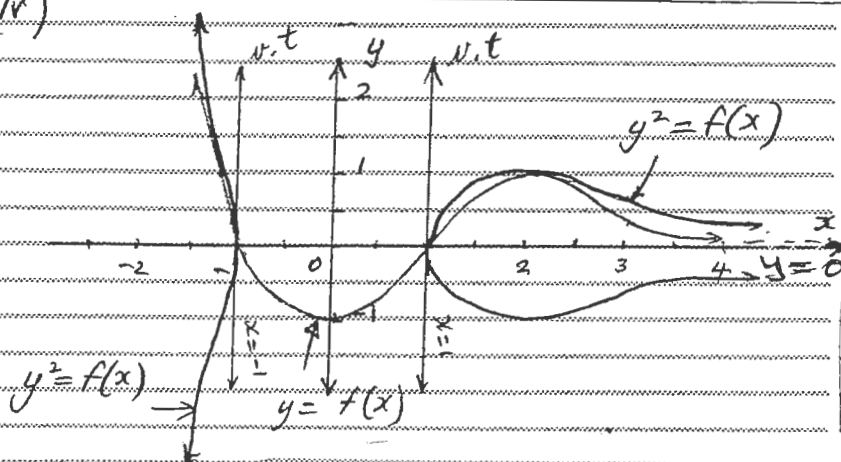
MATHEMATICS Extension 2: Question 2

Suggested Solutions

Marks

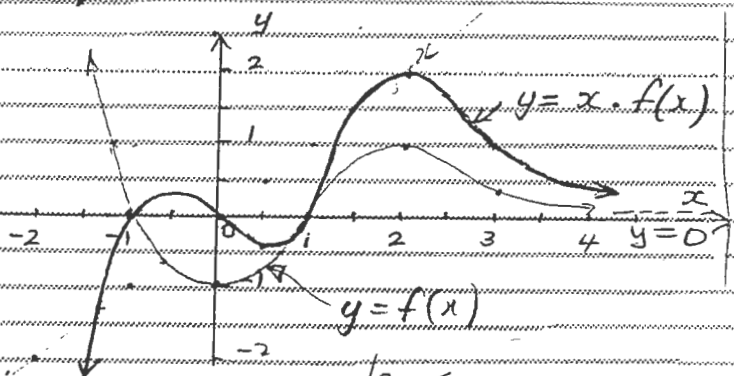
Marker's Comments

(IV)



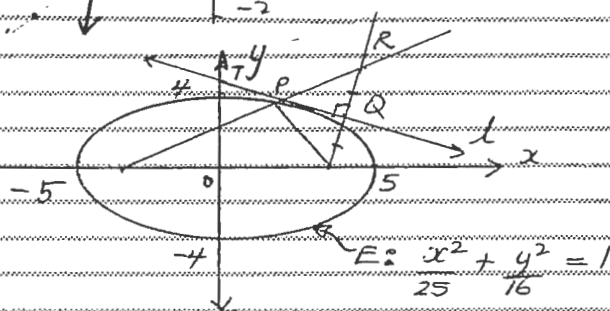
(2)

(V)



(2)

b)



(i) Prove: $SQ = RQ$

Proof: T is the intersection of l, with the y-axis

$\angle TPS = \angle QPS$ (reflection property of an ellipse)

$\angle TPS = \angle QPR$ (vertically opposite angles are equal)

$\therefore \angle QPS = \angle QPR$

In $\triangle SPQ$ and $\triangle RPQ$:-

$\angle QPS = \angle QPR$ (proven above)

$\angle PQR = \angle PQS = 90^\circ$ (as $SQ \perp l$)

PQ is common

$\therefore \triangle SPQ \cong \triangle RPQ$ (AAS)

$\therefore SQ = RQ$ (corresponding sides of congruent triangles are equal)

(1/2) reflection property

(1/2) vertically opposite angles

(1) congruent triangles proof

MATHEMATICS Extension 2: Question ... 2

Suggested Solutions

Marks

Marker's Comments

(ii) Explain why $S'R = 10$
 (1) $S'P + PS = 2 \times 5 = 10$ (sum of distances from any point on the ellipse to the foci equals $2a$)
 From (i), $PR = PS$ (corresponding sides of congruent triangles are equal)
 (1) $\therefore S'P + PR = 10$
 Now, $S'R = S'P + PR = 10$ #

①

① must refer to definition of ellipse

① $PR = PS$ & reason

(iii) Several Methods:

Easiest method:

In $\triangle SRS'$, QQ' is the join of the midpoints of 2 sides of the triangle
 $\therefore QQ'$ is parallel to and half of $S'R$
 ie $QQ' = 5$
 $\therefore Q$ lies on the circle, centre $(0,0)$, radius 5 units, ie $x^2 + y^2 = 25$

②

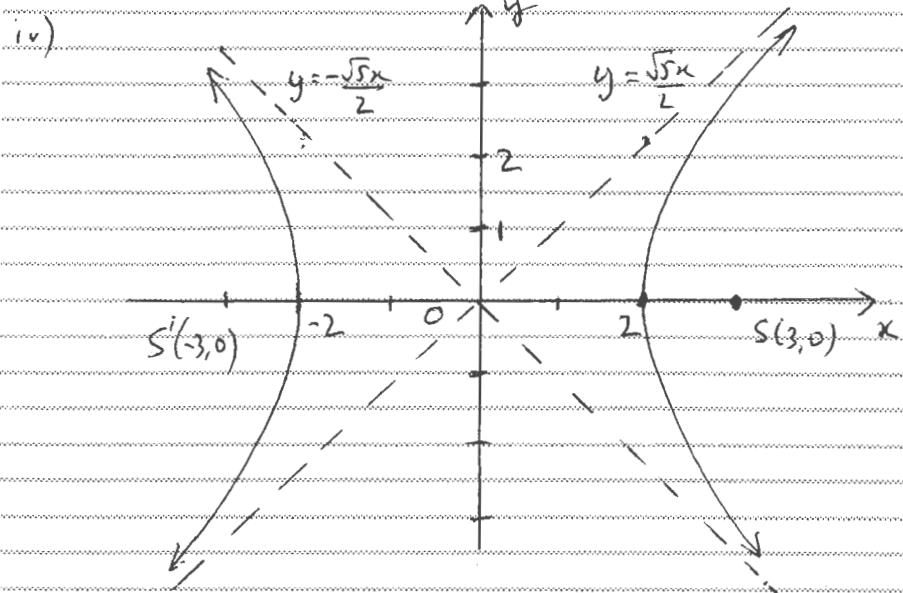
① refer to midpts

① midpoint theorem

① $QQ' = \frac{1}{2} S'R$

① conclusion

2012 TERM 1

MATHEMATICS Extension 1: Question 3
Suggested Solutions
Marks
Marker's Comments


v) $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sqrt{5}\sec^2\theta}{2\sec\theta + \tan\theta} = \frac{\sqrt{5}\operatorname{cosec}\theta}{2}$

Tangent is $(y - \sqrt{5}\tan\theta) = \frac{\sqrt{5}\operatorname{cosec}\theta}{2}(x - 2\sec\theta)$

$$\frac{y}{\sqrt{5}} - \tan\theta = \frac{x}{2\sin\theta} - \frac{1}{\sin\theta\cos\theta}$$

$$\frac{x}{2\sin\theta} + \tan\theta - \frac{y\tan\theta}{\sqrt{5}} = \tan^2\theta - \sec^2\theta$$

$$\frac{x\sec\theta}{2} - \frac{y\tan\theta}{\sqrt{5}} = 1 \quad (\tan^2\theta - \sec^2\theta = -1)$$

vi) At L, substitute $y = x\sqrt{5}/2$

$$\frac{x\sec\theta}{2} - \frac{x\tan\theta\sqrt{5}}{\sqrt{5}\cdot 2} = 1$$

$$x = \frac{2}{\sec\theta - \tan\theta} = 2(\sec\theta + \tan\theta)$$

Substitute for $y \Rightarrow L = (2(\sec\theta + \tan\theta), \sqrt{5}(\sec\theta + \tan\theta))$

Similarly M is $(2(\sec\theta - \tan\theta), \sqrt{5}(\sec\theta - \tan\theta))$

Midpoint ML = $(2\sec\theta, \sqrt{5}\tan\theta)$ from midpoint formula

$\therefore PM = PL$ (P bisects ML)

c) $b^2 = a^2(1 - e^2)$ As $e \rightarrow 0^+$, $b \rightarrow a$ ($a > b > 0$)
 $\therefore$ Shape approaches circle $x^2 + y^2 = a^2$

Best minimum shown for full marks

Many did not put scale on y axis.

2

1

Many made too large a last step without further explanation

1

1 for L and M

1 for midpoint

Finding was penalised.

1

1/2 for circle
1/2 for $a \rightarrow b$ or close

20/2 T/

EXT 2

MATHEMATICS: Question.....4

Suggested Solutions	Marks	Marker's Comments
a) $\frac{dy}{dx} = \frac{-b \cos \theta}{a \sin \theta}$ GRAD of Normal = $\frac{a \sin \theta}{b \cos \theta}$ $y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$ $a \sin \theta x - b \cos \theta y = (a^2 - b^2) \sin \theta \cos \theta$ $\div \cos \theta \sin \theta \quad \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$ but $b^2 = a^2(1 - e^2) = (1 - \frac{1}{3})a^2 = \frac{2}{3}a^2$ $\therefore \frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = \frac{a^2}{3}$	$\frac{1}{2}m$ $\frac{1}{2}m$ $\frac{1}{2}m$ $\frac{1}{2}m$ $\frac{1}{2}m$ $\frac{1}{2}m$	
b i) $\cos \sqrt{ x }$ is even function Area $\hat{=}$ $\int_{-\frac{\pi^2}{4}}^{\frac{\pi^2}{4}} \cos \sqrt{ x } dx = 2 \int_0^{\frac{\pi^2}{4}} \cos \sqrt{x} dx$ ii) $2 \int \cos \sqrt{x} dx = 2 \int \cos u \cdot 2u du \quad u^2 = x$ $2u du = dx$ $A = 2 \times 2 \int_0^{\frac{\pi}{2}} u \cos u du = 4(u \sin u) \Big _0^{\frac{\pi}{2}} - 4 \int_0^{\frac{\pi}{2}} \sin u du$ $= 4 \times \frac{\pi}{2} \times 1 - 0 + 4 \cos u \Big _0^{\frac{\pi}{2}}$ $= 2\pi - 4 \text{ unit}^2$ #	$1m$ $1 + \frac{1}{2}m$ $1 + \frac{1}{2}m$ $1m$	$\cos \sqrt{ x } = \cos x$ even function many students write $\cos \sqrt{x}$ is even = $\frac{1}{2}m$ $\frac{1}{2}m$ for $2u du = dx$ $\frac{1}{2}m$ for limit, Some try to integrate by parts without substitution get a more complicated integral & go nowhere

MATHEMATICS: Question.....4

Suggested Solutions	Marks	Marker's Comments
c i) $I_n = \int_1^e (\ln x)^n dx \quad n = 0, 1, 2, \dots$ $I_n = x(\ln x)^n \Big _1^e - \int_1^e \frac{n(\ln x)^{n-1}}{x} dx$ $= e(\cancel{\ln e})^n - 1(\cancel{\ln 1})^n - n \int_1^e (\ln x)^{n-1} dx$ $= e - 0 - n I_{n-1}$ $= e - n I_{n-1}$	1m 1m	many forgot limit $-\frac{1}{2}n$ must show $(\ln e)^n = 1^n = 1$ $(\ln 1)^n = 0^n = 0$ $-\frac{1}{2}n$ $n(\ln x) \cancel{x} = n(\ln x)$ $-\frac{1}{2}n$ cross out $\frac{x}{x}$ $-\frac{1}{2}n$
ii) $\frac{1}{e}(1 + J_{10}) = \frac{1}{e} \left(1 + \frac{I_{10}}{10!} \right)$ $= \frac{1}{e} \left(1 + \frac{e}{10!} - \frac{10 I_9}{10!} \right)$ $= \frac{1}{e} \left(1 + \frac{e}{10!} - \frac{I_9}{9!} \right)$ $= \frac{1}{e} \left(1 + \frac{e}{10!} - \frac{e - 9 I_8}{9!} \right)$ $= \frac{1}{e} \left(1 + \frac{e}{10!} - \frac{e}{9!} + \frac{I_8}{8!} \right)$ Similarly $= \frac{1}{e} \left(1 + \frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \dots + \frac{e}{2!} - \frac{e}{1!} + \frac{I_0}{0!} \right)$ so $= \frac{1}{e} \left(1 + \frac{e}{10!} - \frac{e}{9!} + \frac{e}{8!} - \dots + \frac{e}{2!} - \frac{e}{1!} + \cancel{\frac{e}{1}} \right)$ $= 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + \frac{1}{8!} - \frac{1}{9!} + \frac{1}{10!}$	$\frac{1}{2}m$ $\frac{1}{2}m$ 1m 1m	must split up $\frac{I_{10}}{10!} = \frac{e}{10!} - \frac{10 I_9}{10!}$ $I_0 = \int_1^e (\ln x)^0 dx = e - 1$ or e $I_1 = \int_1^e (\ln x) dx = 1$ 1m must show last few terms many fudging Alternatively can stop at $I_1 = 1$
iii) $\sum_{r=0}^{\infty} \frac{(-1)^r}{r!} = \lim_{n \rightarrow \infty} \frac{1}{e} [1 + (-1)^n J_n] = \lim_{n \rightarrow \infty} \frac{1}{e} \left[1 + \frac{(-1)^n I_n}{n!} \right]$ $= \left(\frac{1}{e} \right)$	1m 1m	1 + 1m